SLEEP STAGE CLASSIFICATION FROM SINGLE-CHANNEL EOG SIGNALS USING DEEP LEARNING

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1. Abstract

Sleep stage classification is a fundamental task in sleep analysis and plays a crucial role in diagnosing sleep disorders. Traditionally, sleep stage classification is performed manually by experts using polysomnography (PSG) recordings, a time-consuming process subject to variations among analysts. In this project, we propose a deep learning-based approach for the automatic classification of sleep stages using a single-channel electrooculography (EOG-R) signal.

The raw signal is preprocessed using bandpass filtering and calibrated using individual Z-score calibration. The data are segmented into 30-second intervals and classified according to standard sleep stages: wakefulness, Stage 1, Stage 2, Stage 3, and rapid eye movement (REM). We implement a hybrid deep learning architecture combining convolutional neural networks (CNNs) and bidirectional long short-term memory (BiLSTM) to capture both spatial and temporal features.

The model was trained using class balancing techniques and optimized using adaptive rate-learning scheduling and early stop. Experimental results show that the proposed method achieves 80.41% test accuracy, with strong performance in both wakefulness and deep sleep, while highlighting challenges in classifying the first stage.

This work demonstrates the feasibility of using a single-channel electrooculogram (EOG) signal to efficiently and accurately classify sleep stages using deep learning techniques.

2. Introduction

Sleep is a vital physiological process that plays a fundamental role in human health and well-being. Accurate identification of sleep stages is essential for diagnosing various sleep disorders, such as insomnia, sleep apnea, and narcolepsy. Sleep stages are typically identified using polysomnography (PSG), which records multiple physiological signals, including electroencephalography (EEG), electrooculography (EOG), and electromyography (EMG).

Manual scoring of sleep stages by sleep experts is considered the gold standard; however, it is time-consuming and subject to inter-rater variability. Therefore, there is an increasing demand for automated systems capable of reliable and accurate sleep stage classification.

Recent advances in deep learning have significantly improved automated sleep staging performance. Convolutional Neural Networks (CNNs) are effective in extracting local patterns from raw signals, while recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, are well-suited for modeling temporal dependencies in sequential data.

This project focuses on developing an automated sleep stage classification system using a single-channel EOG-R signal. Unlike multi-channel approaches, this method reduces system complexity while maintaining competitive performance. A hybrid CNN–BiLSTM architecture is proposed to effectively learn spatial and temporal features from the signal.

3. Literature Review

Previous studies have explored sleep stage classification using machine learning and deep learning techniques. CNN-based models have demonstrated strong capability in extracting spatial features from physiological signals such as EEG and EOG. In contrast, LSTM networks are widely used to capture temporal dependencies in sequential data.

Many studies have achieved high performance using multi-channel polysomnography (PSG) data; however, these approaches are often computationally expensive. Recently, there has been growing interest in single-channel methods that aim to simplify the system while maintaining acceptable performance.

4. Dataset Description

Source: MESA Sleep Study (NSRR).

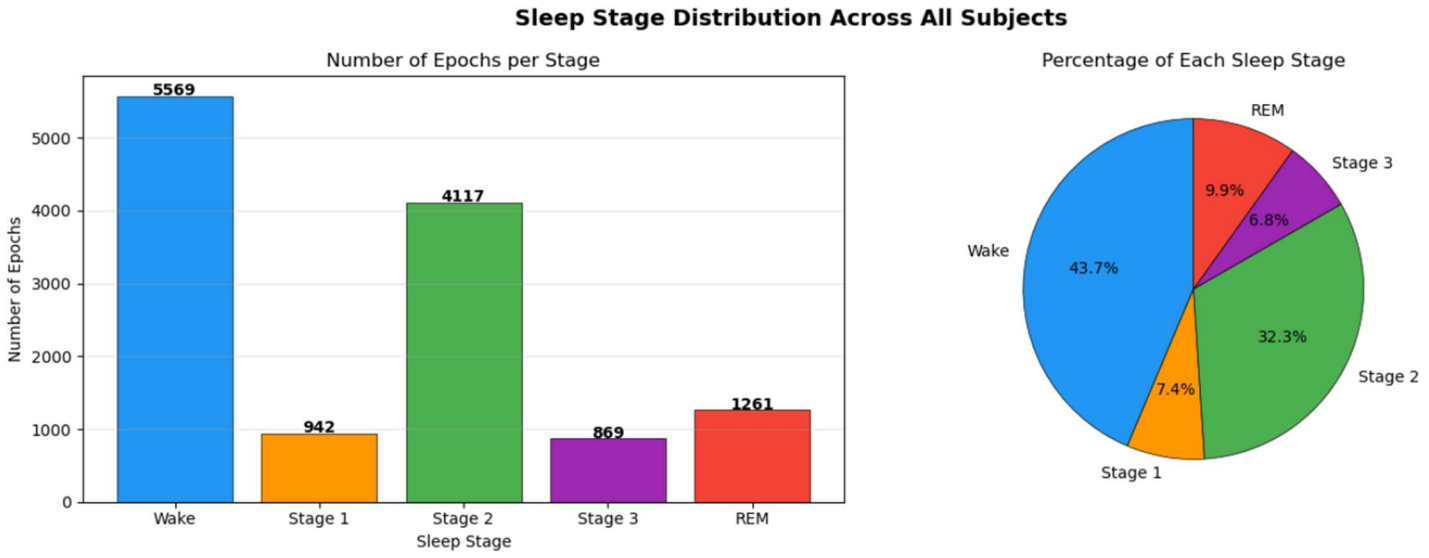
Dataset Type: Polysomnography (PSG).

Number of Subjects: 10.

Total Recording Duration: 106.39 hours.

Sampling Frequency: 256 Hz.

Epoch Length: 30 seconds.




```
Sleep Stage Counts:
  Wake: 5569 epochs (43.7%)
  Stage 1: 942 epochs (7.4%)
  Stage 2: 4117 epochs (32.3%)
  Stage 3: 869 epochs (6.8%)
  REM: 1261 epochs (9.9%)
  Total: 12758 epochs
```

Figure 1: Total Epochs

Selected Channel: EOG-R

Number of Channels in Original Dataset: 27 channels

Sleep Stages

- Wake
- Stage 1
- Stage 2
- Stage 3
- REM

<u>Dataset Structure</u>

Each subject folder contains:

- Signal file (.EDF)

- Annotation file (.XML)The sampling frequency is 256 Hz, and all annotations are based on standardized sleep scoring guidelines.

5. Tools and Libraries

A set of Python libraries was used for data processing and model development.

- **NumPy :** Numerical computations and array handling
- **MNE :** Reading and preprocessing EDF sleep signals
- **SciPy :** Digital signal filtering
- **Scikit-learn :** Data splitting and class weighting
- **TensorFlow :** Deep learning framework
- **Keras :** Model building and training
- **Matplotlib :** Data visualization
- **Seaborn :** Confusion matrix visualization

6. Methodology

➢ 6.1 Signal Preprocessing

In the signal preprocessing stage, a series of steps were implemented to improve data quality and make it suitable for training.

1. **Bandpass Filtering**
   - **Frequency range:** 0.3–35 Hz
   - **Purpose:** Remove noise and irrelevant frequencies
2. **Normalization**
   - **Method:** Z-score normalization
   - Applied individually for each subject

$$z = \frac{\overset{\text{raw score}}{x} - \overset{\text{mean}}{\mu}}{\underset{\text{standard deviation}}{\sigma}}$$

- **Purpose:** Reduce inter-subject signal variation

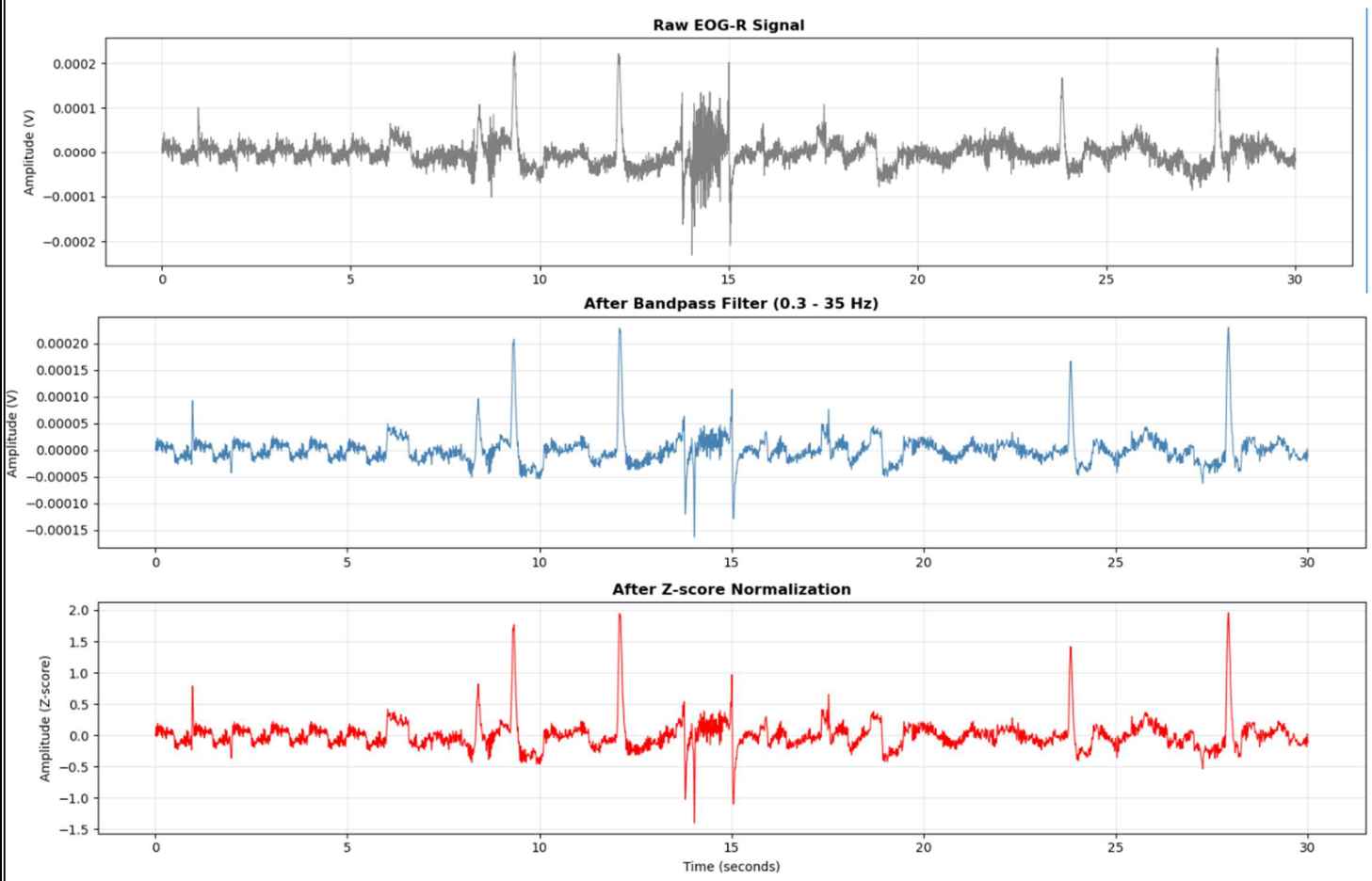


Figure 2:EOG-R Channel: Raw vs Filtered vs Normalized

➢ *6.2 Segmentation*

After preprocessing, the signal was segmented into fixed-length epochs. Each epoch had a duration of 30 seconds, which is the standard duration used in sleep stage analysis.

<u>Segmentation Parameters</u>

- **Epoch Length:** 30 seconds
- **Sampling Frequency:** 256 Hz

- **Segmentation Type:** Non-overlapping epochs

Number of Samples per Epoch

Samples per epoch= 30×256=7680

The signal was divided into consecutive epochs, where each epoch represented one training sample.

Each epoch was then matched with its corresponding annotation label extracted from the XML annotation files.

The sampling frequency was extracted directly from the EDF signal files using the MNE library.

➢ *6.3 Data Preparation*

The data were converted into numerical matrices for training purposes. All time segments were grouped into an X-matrix, and their corresponding labels into a y-matrix.

The data were then divided into three sets:

- Training Set : 70%
- Validation Set : 15%
- Test Set : 15%

Additional Processing

- Stratified splitting
- One-Hot Encoding

6.4 Class Imbalance

Due to the imbalance in the number of samples across the different sleep stages, class weighting was used during training.

Higher weights were assigned to less represented classes, such as Stage 1 and REM, to help the model learn the characteristics of these classes better and reduce bias towards more prevalent classes like Stage 2 and Wake.

This approach contributes to improved model performance, especially in challenging classes.

6.5 Model Architecture

A hybrid model was designed, combining Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), particularly LSTM networks, to leverage the capabilities of each in extracting features from time-series signals.

Conv1D layers are used to extract local features from the signal, such as short-term time patterns, while the Bidirectional LSTM layer is used to capture long-term time relationships between segments.

The model consists of the following stages:

1. **Input Layer:**

This layer receives the signal as time segments of length (number of samples x 1).

2. **Convolution Layers:**

Three consecutive Conv1D layers with ReLU activation functions were used to progressively extract features from the signal.

3. **Batch Normalization:**

Added after each Convolutional layer to improve training stability and accelerate learning.

4. **Max Pooling:**

Used to reduce data dimensions and focus on key features.

5. **Bidirectional LSTM:**

Used to process the temporal dependencies from both the forward and backward directions, helping to better understand temporal relationships.

6. **Dropout:**

Used to reduce overfitting and improve the model's generalization ability.

7. **Fully Connected Layers:**

Dense layers to extract the final representation before classification.

8. **Output Layer:**

A final layer using Softmax to classify the data into five categories representing sleep stages.The model was trained using a Categorical Cross-Entropy loss function and the Adam optimizer, with techniques such as Early Stopping and ReduceLROnPlateau employed to improve performance.

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 7680, 64)	3,264
batch_normalization (BatchNormalization)	(None, 7680, 64)	256
max_pooling1d (MaxPooling1D)	(None, 960, 64)	0
conv1d_1 (Conv1D)	(None, 960, 128)	123,008
batch_normalization_1 (BatchNormalization)	(None, 960, 128)	512
max_pooling1d_1 (MaxPooling1D)	(None, 120, 128)	0
conv1d_2 (Conv1D)	(None, 120, 128)	82,048
batch_normalization_2 (BatchNormalization)	(None, 120, 128)	512
max_pooling1d_2 (MaxPooling1D)	(None, 30, 128)	0
bidirectional (Bidirectional)	(None, 128)	98,816
dropout (Dropout)	(None, 128)	0
dense (Dense)	(None, 64)	8,256
dense_1 (Dense)	(None, 5)	325

Table 1: Model ArchitectureDetails

➢ *6.6 Hyperparameters*

- **Optimizer:** Adam
- **Learning rate:** 0.001
- **Batch size:** 64
- **Epochs:** 30
- **Loss function:** Categorical Cross-Entropy

➢ *6.7 Training Strategy*

The model was trained using the Training Set, with the Validation Set used to monitor performance during the training process.

The Adam (Adaptive Moment Estimation) optimization with an initial learning rate of 0.001 was used because it strikes a good balance between learning speed and stability.

A Categorical Cross-Entropy loss function was adopted because the problem involves classifying multiple categories.

To optimize the training process and prevent overfitting, the following techniques were used:

- **Early Stopping:**

Training is automatically stopped if the model's performance on the Validation Set does not improve after a certain number of epochs, and the best achieved weights are retrieved.

- **ReduceLROnPlateau:**

The learning rate is automatically reduced when the model's performance is stable, helping to optimize the solution.

A batch size of 64 and up to 30 epochs were used, along with class weights, to address data imbalance.

7. Results and Discussion

The model achieved a final test accuracy of 80.41%, demonstrating its ability to classify sleep stages using a single EOG-R channel.

Strong performance was observed in detecting Wake and REM stages due to the clear eye movement patterns captured by the

EOG signal. REM sleep is characterized by rapid eye movements, while Wake stages show higher ocular activity compared to deep sleep stages.

Lower performance was observed for Stage 1 due to its transitional nature and lack of distinct EOG characteristics. Additionally, confusion between Stage 2 and Stage 3 was observed, as these stages are primarily distinguished by EEG-based delta wave activity, which is not captured by EOG signals.

Overall, the results indicate that a single-channel approach can achieve reasonable performance, although it has limitations in distinguishing deeper sleep stages.

7.1 EOG-R Results

```
60/60 ━━━━━━━━━━━━━━━━━━━━ 16s 260ms/step - accuracy: 0.8041 - loss: 0.4671
Final Accuracy: 80.41%
60/60 ━━━━━━━━━━━━━━━━━━━━ 17s 268ms/step
              precision    recall  f1-score   support

        Wake       0.94      0.94      0.94       836
     Stage 1       0.34      0.67      0.45       141
     Stage 2       0.85      0.72      0.78       618
     Stage 3       0.77      0.66      0.71       130
         REM       0.80      0.65      0.72       189

    accuracy                           0.80      1914
   macro avg       0.74      0.73      0.72      1914
weighted avg       0.84      0.80      0.81      1914
```

Figure 3: Classification Performance Metrics for Each Sleep Stage

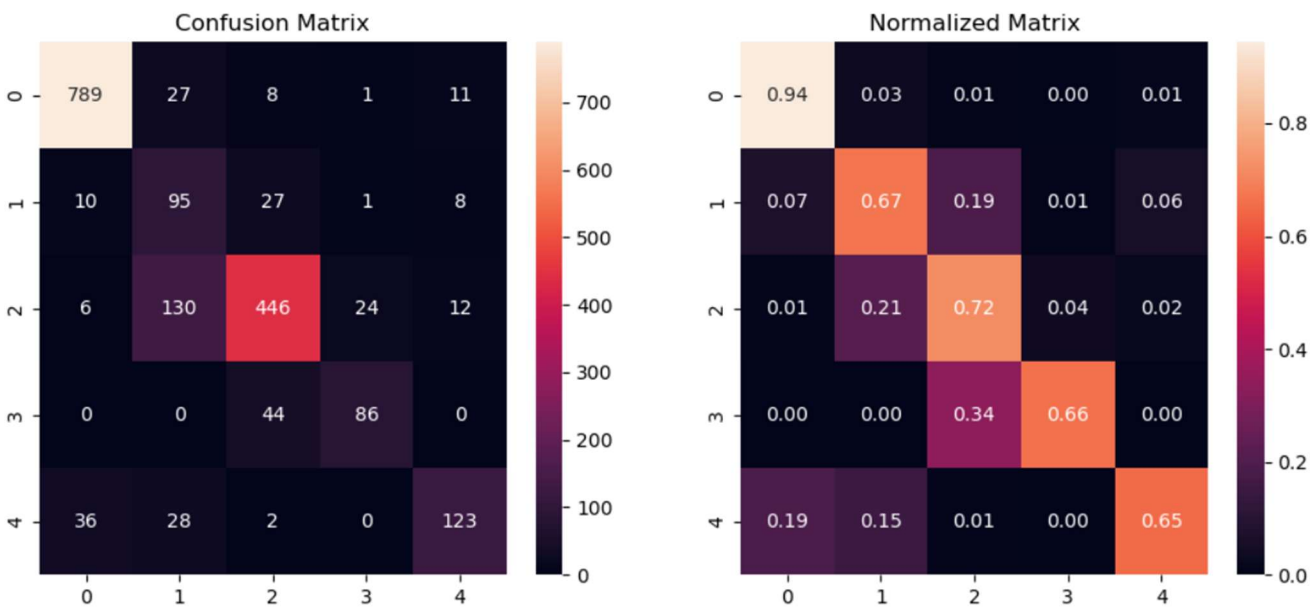


Figure 4: Confusion Matrix of Sleep Stage Classification

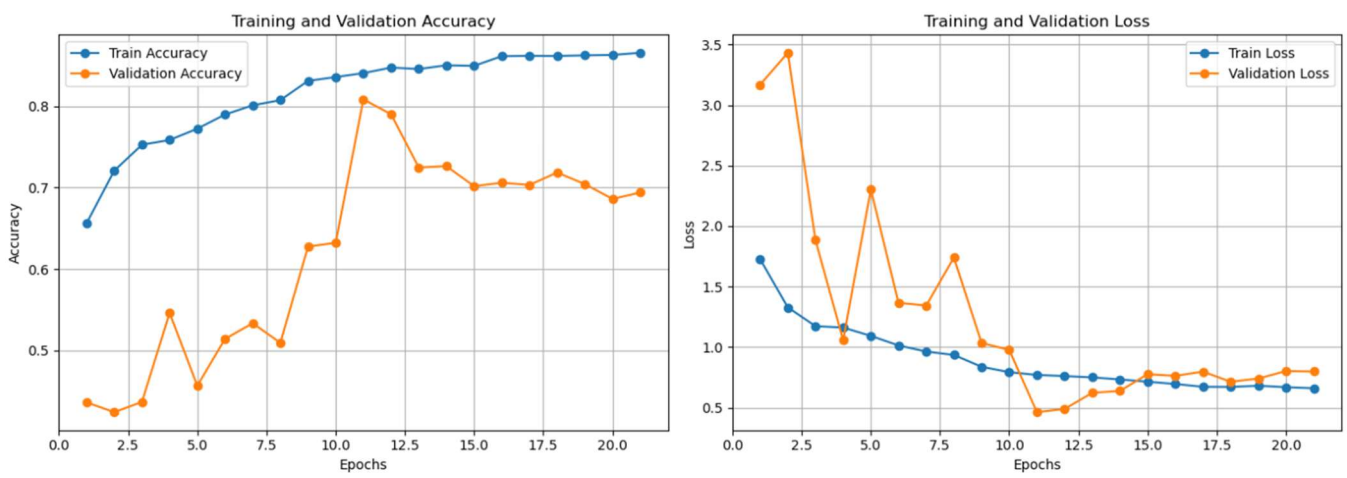


Figure 4: Training and Validation Accuracy and Loss Curves

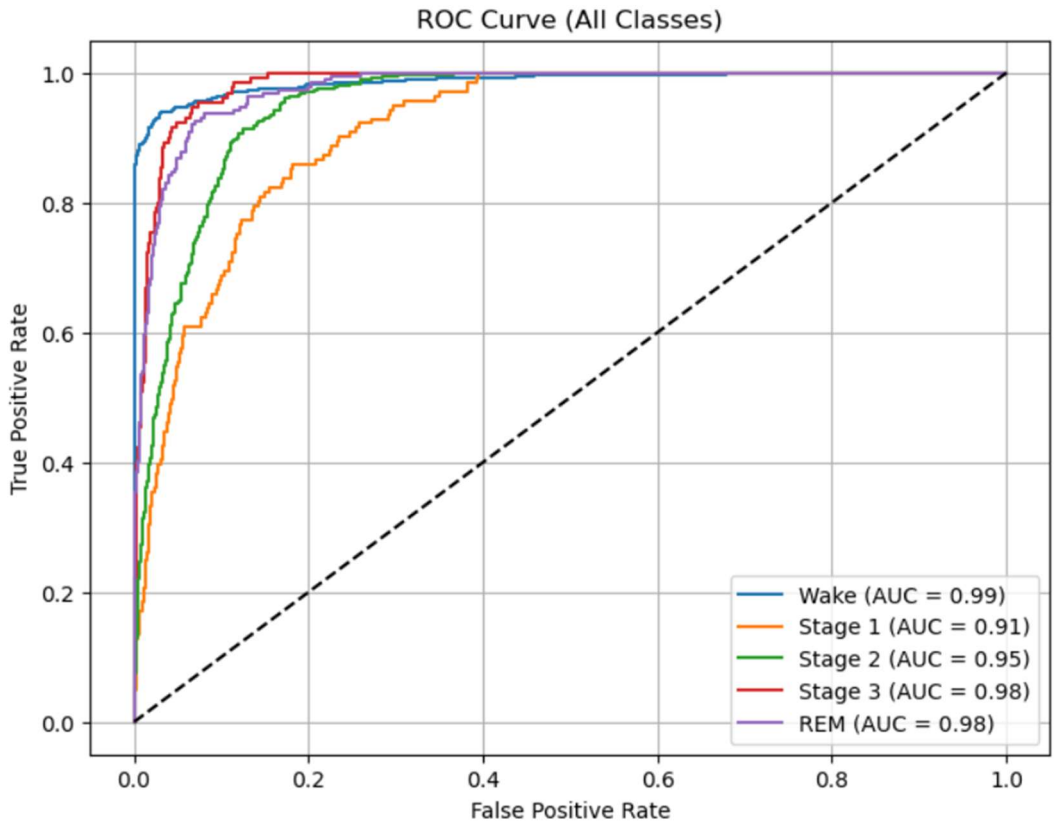


Figure 5: ROC Curve

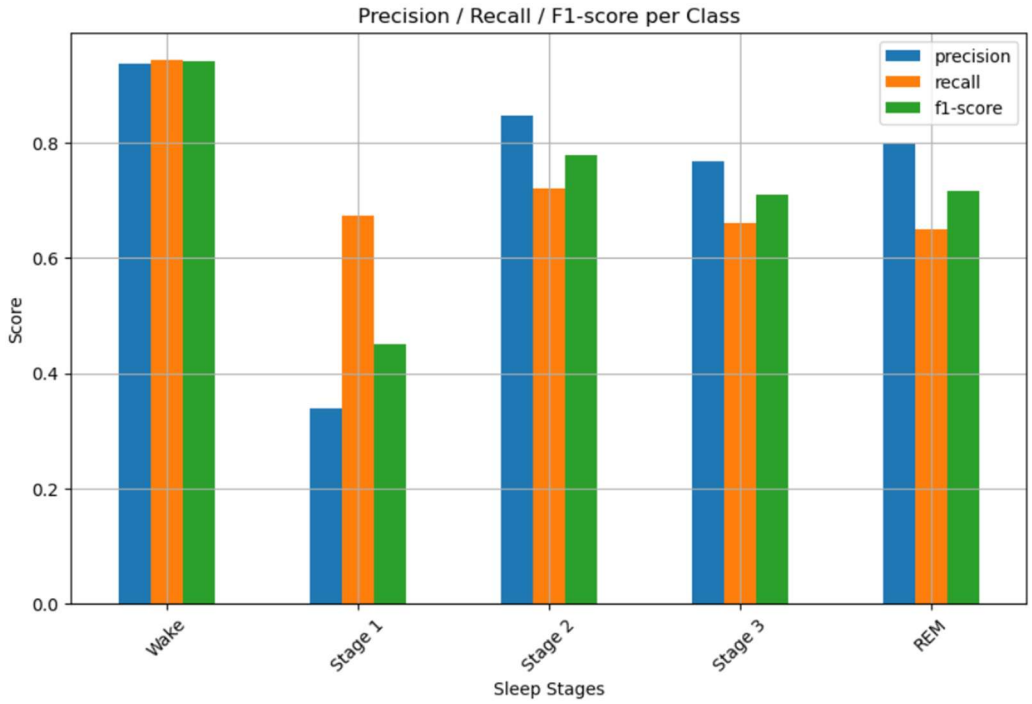


Figure 6:Precision / Recall / F1 Visualization

7.2 Channel Comparison (EOG-R vs EOG-L)

To further validate the channel selection, an additional experiment was conducted using the left EOG channel (EOG-L) under identical preprocessing and model settings.

EOG-R achieved a higher overall accuracy of 80.41% compared to 76.70% for EOG-L. Notable differences were observed in Stage 3 and REM classification, where EOG-R outperformed EOG-L by 0.21 and 0.07 F1-score respectively. These results confirm that EOG-R is the more informative channel for sleep staging, consistent with AASM recommendations.

```
60/60 ━━━━━━━━━━━━━━━━━━━━ 15s 247ms/step - accuracy: 0.7670 - loss: 0.5831
Final Accuracy: 76.70%
60/60 ━━━━━━━━━━━━━━━━━━━━ 17s 261ms/step
              precision    recall  f1-score   support

        Wake       1.00      0.82      0.90       836
     Stage 1       0.31      0.74      0.44       141
     Stage 2       0.81      0.80      0.80       618
     Stage 3       0.85      0.35      0.50       130
         REM       0.60      0.71      0.65       189

    accuracy                           0.77      1914
   macro avg       0.71      0.69      0.66      1914
weighted avg       0.84      0.77      0.78      1914
```

Figure 7: Classification Performance Metrics for Each Sleep Stage

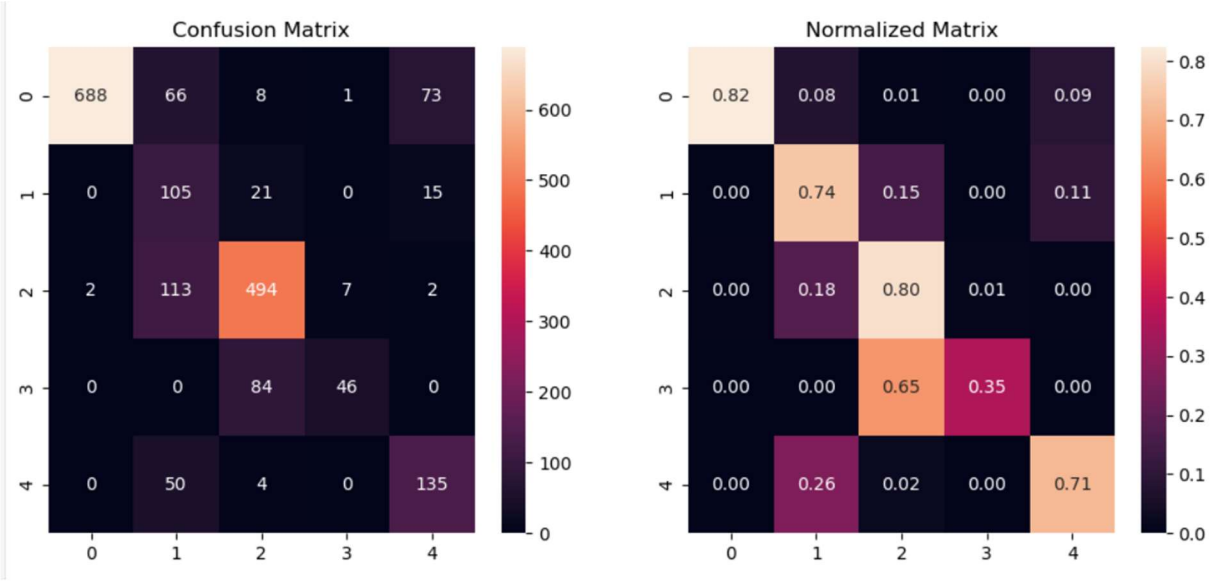


Figure 8: Confusion Matrix of Sleep Stage Classification

8. Conclusion

A sleep stage classification system was developed using a single-channel EOG-R signal with a deep learning approach. The proposed model combines CNN and BiLSTM networks to extract spatial and temporal features from 30-second sleep epochs.

The system was evaluated on a polysomnography dataset consisting of 10 subjects with a total duration of approximately 106.39 hours. The model achieved an accuracy of 80.41% on the test set.

The results demonstrate strong performance in classifying Wake and REM stages. However, challenges remain in distinguishing Stage 1 and differentiating between Stage 2 and Stage 3 due to the limitations of using a single EOG channel.

Overall, the study shows that a simplified single-channel deep learning approach can achieve competitive performance while reducing system complexity.